

XIAOXUAN ZHAO

PhD Researcher in Human–Computer Interaction | AI | XR | Multisensory Interaction
Dipartimento delle Arti, Università di Bologna
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My research investigates how multimodal AI and multisensory XR can enhance design communication across cultural and disciplinary boundaries. Drawing on a background in design practice, I design and experimentally evaluate interactive systems that integrate AI-driven dialogue and sensory-rich XR experiences to support more inclusive and expressive collaboration.

EDUCATION

- Ph.D. in Arts, History, Society (INF/01 – Computer Science)** **Nov 2025 – Present**
Università di Bologna
Thesis: *Enhancing Design Communication through Multimodal AI and Multisensory XR*
Supervisor: Prof. Gustavo Marfia · **Co-supervisor:** Dr. Pasquale Cascarano
CSC Scholarship
- M.A. in Design** **Aug 2022 – Jun 2025**
Beijing Institute of Technology · GPA: 3.74/4.0
Thesis: *Cultural Service System Design for Skill-Based Intangible Cultural Heritage: A Case Study of Yongchang Juci*
Advisor: Prof. Yue Qiu
National Scholarship (Top 0.2%)
- Exchange Program** **Feb 2023 – Jun 2023**
Politecnico di Milano
- B.Eng. in Industrial Design** **Aug 2018 – Jun 2022**
Hohai University · GPA: 4.7/5.0
Outstanding Graduate (Sole Recipient in Major)

PUBLICATIONS

- M. Lai, **X. Zhao**, W. Li, S. Hajahmadi, P. Cascarano, and G. Marfia, “Olfactory Context Cues for Procedural Skill Transfer from Digital Twin-Based Mixed Reality Learning to Real-World Execution,” *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, accepted for publication, 2026. To be presented at *ISMAR 2026*, Bari, Italy.
- X. Zhao**, M. Lai, W. Li, S. Li, P. Cascarano, and G. Marfia, “Bridging the Expressiveness Gap in Cross-Cultural Co-Creation Through AI-Augmented Extended Reality,” *IEEE VRW 2026*, Daegu, Republic of Korea, pp. 519–523.
- W. Li, S. Li, **X. Zhao**, M. Lai, P. Cascarano, and G. Marfia, “Beyond Fashion Vocabulary: Spatial Style Navigation Through Dual-Layer Style-Structural Mapping in XR,” *IEEE VRW 2026*, Daegu, Republic of Korea, pp. 524–528.
- M. Lai, **X. Zhao**, P. Cascarano, and G. Marfia, “Beyond Functionality: Olfactory Cues for Affective Accessibility in Collaborative Social XR,” *IEEE VRW 2026*, Daegu, Republic of Korea, pp. 463–467.
- X. Zhao** and Y. Qiu, “Insight through Dialogue: A Practical Exploration of AIGC in Cross-Cultural Design Research,” *HCI 2024*, Washington, DC, USA, LNCS, vol. 14702, pp. 388–406. **Best Paper Award**

MANUSCRIPTS UNDER REVIEW

- M. Lai, W. Li, **X. Zhao**, G. Vallasciani, P. Cascarano, and G. Marfia, “Confirm or Change Your Choice? Effects of AI Agent Verbal Cues and Ambient Odor on Choice Revision in a Virtual Fashion Store,” submitted to *IEEE VR 2027*.
- S. Hajahmadi, P. Cascarano, **X. Zhao**, L. Donatiello, and G. Marfia, “Recorded Personalized Avatar Faces as a Re-Identification Surface: A Technical Benchmark and Video-Based Privacy–Utility Study,” submitted to *IEEE VR 2027*.

RESEARCH EXPERIENCE

Multisensory XR for Design Communication and Skill Transfer

Nov 2025 – Present

Università di Bologna · Supervisor: Prof. Gustavo Marfia · Co-supervisor: Dr. Pasquale Cascarano

Investigating how olfactory cues shape user behavior in XR environments across two studies: scent-context reinstatement for procedural skill transfer in MR, and olfactory context congruency with AI verbal cues in VR fashion retail; designing experimental protocols, leading data collection, and authoring methodology sections.

CrossProbe: AI-Assisted Cross-Cultural Design Research

Oct 2023 – Jun 2025

Beijing Institute of Technology · Advisor: Prof. Yue Qiu

Explored how multi-agent AI systems can augment cross-cultural design research by replacing conventional surveys with natural conversational interactions. Designed the platform, led comparative experiments and user studies evaluating efficiency and response quality.

PRESENTATIONS

IEEE VR 2026 Workshop on SIC-XR

Mar 2026

“Bridging the Expressiveness Gap in Cross-Cultural Co-Creation Through AI-Augmented Extended Reality”

Daegu, Republic of Korea

HCI 2024

Jun 2024

“Insight through Dialogue: A Practical Exploration of AIGC in Cross-Cultural Design Research”

Washington, DC, USA

TEACHING EXPERIENCE

Teaching Assistant, AI Design Innovation

Jun 2024 – Aug 2024

Hangzhou Shenke Creative Design Co., Ltd.

Facilitated case-based learning on design methods and AI-assisted design for 21 students, mentoring 4 team projects submitted to design competitions.

Teaching Assistant, Integrated Innovative Design

Mar 2024 – May 2024

Beijing Institute of Technology · Prof. Nan Yang

Prepared instructional materials and delivered lectures on integrating generative AI into design practice.

PROFESSIONAL SERVICE

Reviewer, IEEE ISMAR Demo Track

2026

DESIGN AWARDS

Golden Pin Concept Design Award — *Best Concept Design Nominee*, Top 0.32% (2024)

European Product Design Award — *Winner* (2024)

Spark Design Awards — *Gold Winner* (2024)

LANGUAGES

Chinese (native)

English (fluent)